



Synergistic effect of ACC deaminase producing *Pseudomonas* sp. TR15a and siderophore producing *Bacillus aerophilus* TR15c for enhanced growth and copper accumulation in *Helianthus annuus* L

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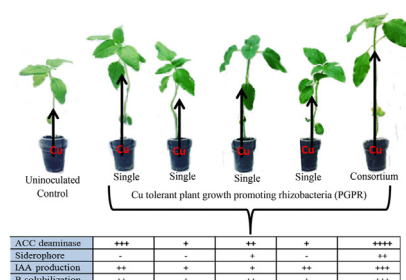
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HIGHLIGHTS

- *Pseudomonas* sp. TR15a and *Bacillus aerophilus* TR15c produced high ACC deaminase.
- Consortium of TR15a + TR15c showed high resistance to multiple metals and antibiotics.
- Siderophore production and ACC deaminase activity were improved for consortium.
- Growth and Cu concentration in sunflower enhanced after application of consortium.
- Consortium act as better phytostabilizer than single inoculant for Cu remediation.

GRAPHICAL ABSTRACT



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ABSTRACT

Copper (Cu) is an essential element, however its excess into the environment causes detrimental effect on plant and risks for public health. Four Cu and drought tolerant 1-aminocyclopropane-1-carboxylate (ACC) deaminase producing rhizobacteria were isolated from the roots of *Trifolium repens* L. growing on Cu smelter contaminated soils, characterized and identified based on 16S rRNA gene sequencing. A consortium of high ACC deaminase (53.74 μM α -ketobutyrate mg^{-1} protein h^{-1}) producing bacteria *Pseudomonas* sp. strain TR15a + siderophore producing *Bacillus aerophilus* strain TR15c significantly ($p < 0.05$) produced better results for multiple-metal tolerance including Cu (1750 mg kg^{-1}), antibiotic resistance (ampicillin, kanamycin, chloramphenicol, penicillin, tetracycline, and streptomycin) and plant growth promoting attributes (phosphate solubilization: 315 mg L^{-1} , indole-3-acetic acid (IAA) production: 8 mg L^{-1} , ammonia and hydrogen cyanide production) as compared to individual isolates. Pot scale experiment (enriched with 100 mg Cu kg^{-1}) showed inoculation of *Helianthus annuus* seeds with consortium of TR15a + TR15c had significantly ($p < 0.05$) improved seed germination by 32%, total dry biomass by 64%, root Cu by 47% and shoot Cu by 75% as compared to uninoculated control whereas 0.2–7 fold higher results were observed for above stated parameters as compared to four individual isolates

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studied. The result suggests consortium of ACC deaminase producing *Pseudomonas* sp. TR15a and siderophore producing *B. aerophilus* TR15c could play a vital role in enhanced Cu uptake and improvement of biomass and may provide a better alternative for decontamination of Cu contaminated natural ecosystem than individual isolates.

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1. Introduction

Copper (Cu) is an essential element for growth of both plants and human being however, its excess in concentration could lead to toxic effect on both organisms. Cu gets easily introduced into the environment by natural and/or anthropogenic means via mining, smelting, sludge, leaching, soil erosion, fertilizers, municipal compost, fungicides and pesticide, city solid waste; and automobiles (Ma et al., 2009; He et al., 2010). Such excess of Cu in the ecosystem could lead to soil nutrients depletion, vegetation loss and risks to human health (Ma et al., 2009; Zhang et al., 2011). Thus, it is necessary to combat this problem by implementing new and safer technology involving the use of green technology.

Up to recent times, the mostly physical and chemical methods were implemented for decontamination of metal polluted sites however, they were less competent because of big infra-structure requirement, high operational cost and generation of secondary contaminants. In recent times, phytoremediation acquired its significant importance because of its ease in application, eco-friendly nature, and importantly cost effectiveness (Romkens et al., 2002; Kumar et al., 2017; Mukhopadhyay et al., 2017). Big biomass plants are the principle need of phytoremediation and high metal accumulating (hyperaccumulator) plants do not fulfill this requirement and this make it influential to get switched towards microbe-assisted phytoremediation for resolving this global problem (Rajkumar et al., 2009; Zhang et al., 2011; Kumar et al., 2021). Microbe-assisted remediation showed their effectiveness however, single inoculant sometimes found insufficient to show all the beneficial effects and could only produce partial benefits to the plants. These limitations have prompted us to decipher the plant growth promoting rhizobacteria (PGPRs) and possible synergistic effect of bacterial isolates for improved plant growth and metal uptake which are still seldom studied.

Rhizosphere of the metal tolerant plant harbors PGPRs which can enhance plant growth through phyto-hormone production, nitrogen fixation, mineral solubilization, nutrient and improves plant resistance to pests and diseases (Singh et al., 2015; Gange and Gadhave, 2018; Grobelak et al., 2018). PGPRs alleviates stress by secreting 1-aminocyclopropane-1-carboxylate (ACC) deaminase whereas siderophores production helps in iron-chelation complexation. The ACC deaminase is an enzyme which cleaves ACC (a precursor of ethylene in plants) into ammonia and α -keto-butyrate (Glick 2014; Nascimento et al., 2018; Chandra et al., 2019; Gupta and Pandey, 2019) and protects plants against flooding, drought, salt, flower wilting, metals, organic contaminants, both bacterial and fungal pathogens and other environmental stress factors. Glick (2014) also reported that ACC deaminase producing bacteria can promote plant growth and can help to feed the world. Siderophore are iron-chelating complex which helps in providing nutrient and enhancing the bioavailability of the metals for enhanced phytoaccumulation. Along with the other metals like Fe, Zn, siderophore can easily solubilize the Cu and upsurges its bioavailability for improved Cu-extraction by forming a stable complex with it (Dimkpa et al., 2009; Rajkumar et al., 2010; Hesse et al., 2018; Kumar et al., 2021). Single PGPR assisted

phytoremediation is well studied (Carrillo-Castaneda et al., 2005; Ma et al., 2013; Singh and Jha, 2018), however, very few studies are reported for decontamination of Cu polluted soil using bacterial consortium, and no consortium studies for Cu tolerant ACC deaminase and siderophore producing rhizobacteria.

Sunflower (*Helianthus annuus* L.) a bioenergy hyperaccumulator plant of Cd also showed significant potential to accumulate various metals including Cu (Li and Ramakrishna, 2011; Kolbas et al., 2015). Because of high economic value, big biomass and extensive root system with multiple metal accumulating potential, it became one of the best choices for phytoremediation of metal polluted derelict sites. A fourteen-year research review by Vamerali et al. (2010) had also reported sunflower as second most frequently used field crop in phytoremediation. Implementation of most of the Cu-phytoremediation is mainly restricted to Brassicaceae and Fabaceae family plants and thus there is a great need to explore the potential of this Asteraceae family plant with rhizobacteria for improved results.

The main objectives of the present study were to isolate and characterize the Cu-tolerance, drought and antibiotic resistance of rhizospheric bacteria from the root tissue of Cu smelter influenced soil, to study their plant growth promoting (PGP) attributes mainly ACC deaminase and siderophore production, and to elucidate synergistic effect of PGPRs on growth and Cu accumulation in sunflower plant for clean-up of Cu contaminated sites.

2. Materials and methods

2.1. Soil sampling and chemical characterization

The rhizospheric soil was collected from one of the most dominant plant species *Trifolium repens* L. growing on Cu smelter contaminated site of Karabash, Chelyabinsk oblast, Russia. Samples were collected during the blooming period (June' 2016) with a mean temperature of 20 °C. The matured plants were carefully uprooted to prevent damage of roots and shaken in a polyethylene zip bag to collect adhered soil particles. The samples were transferred to the laboratory, air dried, homogenized, passed through sieve (<2 mm) and reserved for chemical analyses. The rhizospheric soil adhered on the root surface were brushed out, collected in a separate sterile zip bags and stored at 4 °C in a refrigerator for isolation of rhizobacteria. However, no nodules were observed even in the matured plants which possibly because of the very high multiple metal contamination (Dary et al., 2010).

The pH (1:2.5; w/v) was measured in soil-water solution by digital pH meter (Waterproof Pocket pH tester, Hanna Instruments, Germany). Organic carbon (OC) was determined by rapid dichromate oxidation method (Walkley and Black, 1934). Total metal concentrations (Ca, Mg, Cu, Cr, Cd, Pb and Ni) in soil samples were determined by digesting 0.25 g of soil with a mixture of HNO₃:HClO₄:HF (5:1:1, v/v/v) on a hot plate. The samples were filtered using filter paper, diluted and analyzed using ICP-AES (Inductively coupled plasma-atomic emission spectrometer, iCAP 6500 Duo, Thermo Fisher, USA).

2.2. Isolation of rhizobacteria

For isolation of rhizobacteria the method of [Ma et al. \(2015\)](#) was adopted and slightly modified. About 10 g of rhizospheric soil was mixed with 90 mL of phosphate buffer (pH 6.5) and shaken in an orbital shaker (Biosan, Latvia) at 180 rpm for 2 h at 28 °C. The sample was serially diluted and 100 µL of sample was spread on Luria-Bertani (LB) agar plate supplemented with 100 mg Cu L⁻¹ and 75 mg L⁻¹ of cyclohexamide to inhibit the fungal growth. For the growth of culturable bacteria, the plates were incubated for next 3 days at 28 °C in bacterial incubator (Remi, India). A total of 26 different bacterial colonies were selected based on morphology and appearance. The isolates were further purified and subjected to minimum inhibitory concentration (MIC) and antibiotic resistance test.

2.3. Minimum inhibitory concentration (MIC), drought and antibiotic resistance

The bacterial isolates were subjected with increasing metal concentrations to determine their MIC at which their growth arrests (for Cu: 250–1750 mg kg⁻¹; Cd: 50–750 mg kg⁻¹; Cr: 100–1000 mg kg⁻¹; Pb: 500–3500 mg kg⁻¹ and Ni: 250–2250 mg kg⁻¹) and allowed to grow on LB agar plate for next 5 days at 28 °C. To prevent the precipitation of high concentration of metal, the agar plates were supplemented with EDTA (ethylene diaminetetraacetic acid). Simultaneously, another set of LB agar plates without metal was prepared by adding EDTA to counter check its effect on the bacterial growth. Plates without EDTA and metal were considered as control.

The ability of Cu tolerant rhizobacteria to tolerate drought stress was assessed by growing them in polyethylene glycol (PEG) 6000-infused tryptone yeast-extract and glucose (TYEG) agar medium. The growth of bacterial strains after 3 days under different water potentials ranging between mild (−0.05 MPa, 5% PEG 6000) to over moderate (−0.73 MPa, 25% PEG 6000) were observed and compared with control ([Ma et al., 2016](#); [Kumar et al., 2021](#)).

Antibiotic resistance of the bacteria was checked by disc diffusion method ([Rajkumar et al., 2013](#)). An overnight grown bacterial culture (approximately 10⁸ cfu mL⁻¹) in LB broth was plated (10 µL) on LB agar plate followed by placing antibiotic discs of 6 mm on agar plate by sterile forceps and kept in a bacterial incubator at 28 ± 2 °C for 3 days. The concentrations of the antibiotic disc used were ampicillin (10 µg), tetracycline (30 µg), streptomycin (30 µg), penicillin (6 µg), chloramphenicol (30 µg) and kanamycin (30 µg). The zone of inhibition around the disc was measured and categorized under as resistant, intermediate and sensitive on the basis of company sensitivity guidelines (NICF, Russia).

2.4. Bacterial identification and phylogenetic analysis

The isolates showing high metal tolerance were identified based on morphology and 16S rRNA gene sequencing. The genomic DNA was isolated using DNeasy Plant Mini Kit (Qiagen, Germany). Further purification was done by NucleoSpin gDNA clean-up kit (Macherey-Nagel, Germany). The 16S rRNA gene was amplified by polymerase chain reaction (PCR) by optimizing with Maxima Hot Start Green PCR Master Mix (Thermo Fisher Scientific) and universal primers 27F (5′-GAGTTTGATCACTGGCTCAG-3′) and 1492R (5′-TACGGCTACCTTGTTACGACTT-3′) and sequenced by Macrogen Inc., Netherlands ([Kumar et al., 2021](#)). Chromas software was used for sequence analysis and BLASTn algorithm was executed to compare with the NCBI sequences (≥99%). The sequences of strain TR15a, TR15b, TR15c and TR15e were submitted in NCBI database

under accession number MW682304, MW682305, MW682306 and MW682307, respectively.

2.5. Plant growth promoting attributes

2.5.1. ACC deaminase activity

A 15 mL of half strength trypticase soya broth (TSB) was inoculated with bacterial strain and allowed to grow at 220 rpm at 28 ± 2 °C until the stationary phase was achieved. The samples were centrifuged at 6000×g for 5 min at 4 °C and pellets were washed twice with 5 mL of Dworkin and Foster (DF) minimal media. The pellets were resuspended in 7.455 mL of DF minimal media along with 45 µL of 0.5 M ACC and incubated at 28 °C for 24 h at 200 rpm. Again, the samples were centrifuged at 4 °C for 10 min at 6000×g and pellets were washed twice with 5 mL of filter sterilized (0.22 µm) 0.1 M Tris buffer (pH 7.6). The pellets were resuspended in 1 mL of 0.1 M Tris buffer (pH 7.6) and 1 mL was transferred in sterile eppendorf tube followed by centrifugation at 16,000×g for 5 min at 4 °C. The pellets were then resuspended in 600 µL of 0.1 M Tris buffer (pH 8.5) with 30 µL of toluene. Suspension was vortex at maximum speed for 30 s and kept on ice box. Now, 200 µL of lysate was transferred in a new eppendorf tube and rest was stored at −20 °C for protein assay. A 20 µL 0.5 M ACC was added, vortex for 5 s, incubated at 30 °C for 15 min and 1 mL 0.56 M HCl was added, centrifuged for 5 min at 16,000×g. Next, 1 mL of supernatant was taken and 800 µL HCl along with 300 µL of reaction solution was added and incubated at 30 °C for 30 min. Control was prepared simultaneously without adding supernatant. Now, 2 mL 2 M NaOH was added and optical density was recorded at 540 nm using UV–Visible spectrophotometer (Teccan, Thermo scientific, USA). α -ketobutyrate was prepared from 0 to 1 mM by adding 1800 µL HCl to 300 µL reaction solution and incubated for 30 °C for 30 min followed by addition of 2000 µL of NaOH. The optical density (OD) was recorded at 540 nm. Determination of protein was done following the method of [Bradford \(1976\)](#). Bovine serum albumin was used to establish a standard curve.

2.5.2. Phosphate solubilization and indole-3-acetic acid (IAA) production

Briefly, bacterial isolates were grown in 50 mL of LB broth medium. A 100 µL bacterial sample (OD was adjusted to 10⁸ cfu mL⁻¹) was inoculated in freshly prepared National Botanical Research Institute's phosphate growth (NBRI-P) medium (pH 7.0 ± 0.2) and incubated for next 10 days at 28 °C. One mL of the inoculum was pipetted out to a fresh eppendorf tube and centrifuged for 6 min at 10,000×g. A 29 µL supernatant was pipetted in a 96-well microtitre plate with 114 µL of double distilled deionized water and 57 µL of Vanado-molybdc reagent. Samples were incubated for 10 min and OD was observed at 420 nm using UV–Visible spectrophotometer (Teccan, Thermo scientific, USA) ([Ribeiro and Cardoso, 2012](#)). Parallely, control samples without bacterial inoculum was also checked. To calculate the concentration of phosphate produced, calibration curve was prepared between 0 and 500 mg L⁻¹ using KH₂PO₄ and H₂SO₄ (10 N) in double distilled deionized Millipore water.

The IAA production was measured by slightly modifying the method of [Ma et al. \(2015\)](#). Freshly prepared overnight grown bacteria in LB broth supplemented with 200 µL of L-Tryptophan (at 28 °C) were centrifuged at 10,000×g for 6 min. Supernatant (100 µL) was mixed with Salkowski's reagent (200 µL) (49 mL of 35% HClO₄ and 1 mL of 0.5 mol L⁻¹ FeCl₃), incubated in dark for 30 min and OD was measured at 530 nm in 96-well multi-titre plate in UV–Visible spectrophotometer (Teccan, Thermo scientific, USA). IAA concentration was measured by preparing a calibration curve ranging between 0 and 25 mg L⁻¹ using commercially

available IAA (Sigma-Aldrich).

2.5.3. Siderophore, ammonia and hydrogen cyanide production

Siderophore production was assessed by following method of Schwyn and Neiland (1987). Briefly, Fe-CAS indicator solution (100 mL) was mixed with principle buffer (800 mL) followed by mixing with 70 mL of auxiliary buffer. Lastly, the prepared solution was mixed with casamino acid solution, plated and inoculated with $5 \mu\text{L}$ of 10^8 cfu mL^{-1} containing rhizobacteria. The blue agar plates were incubated at $28 \pm 2^\circ\text{C}$ for 7 days. Halo zones of orange to yellow around bacterial colony was measured (mm) and recognized as siderophore producing bacteria.

Ammonia production and hydrogen cyanide production was observed in broth and in agar plates by following the method of Kshetri et al. (2018) and Sharma et al. (2018), respectively.

2.6. Seed sterilization and plant growth experiment

The seeds similar in size and shape of *Helianthus annuus* L. (variety Lakomka) were sorted to conduct the experiment. The surface sterilization of seeds was performed by following the method of Kumar et al. (2021).

Non-anthropogenic influenced soil were collected from the botanical garden of Ural Federal University, Ekaterinburg, hand sorted to remove stones and litters, homogenized and sieved through $<2 \text{ mm}$ followed by autoclaving at $130 \pm 3^\circ\text{C}$ for 1 h in an autoclave for three consecutive days to kill microbial content of substrate. Final concentration of $100 \text{ mg Cu kg}^{-1}$ was achieved by spiking the soil with $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$. The 30 g of sterile clay stones were placed at the bottom of pots for providing good aeration followed by 300 g of sterile Cu-spiked soil and was irrigated with 200 mL of distilled water. Pots were left for equilibration in an open chamber and next 15 days. Freshly, prepared overnight grown bacterial cultures on LB medium were centrifuged at $8000 \times g$ for 10 min, pellets were washed multiple times with sterile tap water to remove traces of media and other impurities. The concentration of bacterial suspension, single or consortium was adjusted to OD 1.0 (10^8 cfu mL^{-1}) and 5 mL of bacterial inoculum was added in the 3 days old seedling containing pots and allowed to grow for another 25 days. For preparation of consortium, OD of each rhizobacteria was adjusted to ~ 0.5 and was mixed together to produce a final OD of 1.0 (10^8 cfu mL^{-1}). In control, 5 mL of sterile distilled water was added in place of rhizobacteria. After 25 days of growth period, plants were removed carefully from the pots, washed multiple times in running tap water and segregated into root and shoot. Ultrasonication chamber (Analab, India) was used for complete removal of adhered soil particles from the roots. Post-harvest soil was analyzed to check decrease in Cu content. For determination of bacterial root and soil colonization, sunflower roots with adhered soil were soaked in LB broth and left on shaker for 60 min to recover the bacteria. The number of bacteria surviving in plant root rhizosphere in LB agar with 100 mg Cu L^{-1} and $10 \mu\text{g mL}^{-1}$ of ampicillin was calculated by plate dilution technique (Li and Ramakrishna, 2011). The length of plant tissues was measured after drying with paper towel, tissues were oven dried at 80°C until constant weight and biomass was measured. Accurately weighted 100 mg of sample was digested in HNO_3 using microwave digestion system (CEM MARS 5, NC, USA) using conc. HNO_3 and concentration of Cu was analyzed using ICP-AES.

2.7. Statistical analyses

Each sample was done in three replicates unless and until specified and errors were presented in standard deviation (SD). The optical density of 10^8 cfu mL^{-1} was used for performing all

experiments unless and until specified. The normality (Shapiro Wilk test) and homogeneity of variances (Levene's test) were tested and significant difference between the multiple-variables was determined by one way ANOVA. Kruskal Wallis test were used to analyze non-normal data. Different alphabetical letter indicates significant difference at $p < 0.05$. All the reagents were of analytical grade and samples were prepared using double deionized Millipore water. The calibration coefficients were maintained at not less than 0.999. The standard reference material (JSC Ural Plant of Chemical reagent, Russia) and blank reagent was used for calibration, and precision and accuracy of the method was ascertained by replicate analysis of the analyzed elements.

3. Results and discussion

3.1. Chemical characterization and heavy metal concentration in rhizospheric soil

The pH of the soil was found near to neutral with low content of organic carbon and high Mg:Ca ratio. Concentration of Cu was maximum followed by $\text{Ni} > \text{Pb} > \text{Cr} > \text{Cd}$. Presence of high Mg, Ni and Cr showed serpentine nature of the soil whereas very high Cu content was due to geogenic origin and influence of active century old Cu smelter (Table 1).

3.2. Bacterial identification and characterization

Out of twenty-six Cu-resistant strains (100 mg L^{-1}) isolated from the roots of *T. repens*, twelve isolates showed IAA production, eleven solubilized phosphate, two produced siderophore, and five strains showed ACC deaminase activity and were preliminarily identified on the basis of their morphology (appearance, color, texture, shape, growth, elevation, size, pigmentation and optical property), and biochemical tests (data not shown).

Four bacterial isolates were selected on the basis of their higher capability to show at least four PGP attributes (including ACC deaminase) out of five. Bacterial strains possessing multiple PGP traits showed their evidence in improved growth of the plant and metal resistance. In addition, ACC deaminase producing bacteria also showed its significant role in root elongation by reducing ethylene production in stressed plants, hence enhanced plant growth (Belimov et al., 2009; Ma et al., 2015). All the four ACC deaminase producing bacteria were faster in their growth rate, and 75% of them were gram negative which showed positive test for catalase, oxidase, citrate and negative for urease, methyl red and hydrogen sulphide. The 16S rRNA gene sequencing identified the four strains as *Pseudomonas* sp. TR15a, *Pseudomonas* sp. TR15b, *Bacillus aerophilus* TR15c, and *Stenotrophomonas rhizophila* strain TR15e (Table 2). Many authors had previously isolated *Bacillus* and *Pseudomonas* species from the metal contaminated sites and reported their significant role in plant growth promotion (Zhang et al., 2011; Kumar et al., 2012; Grobelak et al., 2018).

3.3. Minimum inhibitory concentration, drought resistance and antibiotic sensitivity test

Microorganisms dwelling in the rhizosphere of naturally growing dominant plants under metal and drought stressed condition often possess various metal and antibiotic resistances and possibly help them to adapt to stressful conditions (Pal et al., 2005; Ma et al., 2015; Chen et al., 2019; Kumar et al., 2021). In the present study, all four PGPRs showed strong and variable tolerance to different metals and were in the order of $\text{Pb} < \text{Ni} < \text{Cu} < \text{Cr} < \text{Cd}$. The maximum tolerance of Cu and Cr were 1750 and 1000 mg L^{-1} , respectively, exhibited by all four strains whereas it varied between

Table 1Chemical characterization and heavy metal concentrations (mg kg⁻¹) in rhizospheric soil of *T. repens* (Mean $\pm$ SD; n = 3).

Parameters	Mean $\pm$ SD	(Min–Max)	Metals	Mean $\pm$ SD	(Min–Max)
pH (1:2.5; w/v)	5.58 $\pm$ 0.14	5.74–6.01	Cu	2283.48 $\pm$ 685.70	1754.27–3058.12
Organic carbon (%)	0.42 $\pm$ 0.04	0.390.47	Ni	485.05 $\pm$ 26.12	461.89–513.37
Total Ca (mg kg ⁻¹)	20,109 $\pm$ 809	19,283–20901	Cr	744.24 $\pm$ 196.50	566.79–955.41
Total Mg (mg kg ⁻¹)	57,544 $\pm$ 5168	51,591–60890	Cd	6.84 $\pm$ 0.86	6.02–7.73
Mg:Ca	2.86	2.68–2.91	Pb	904.04 $\pm$ 118.03	806.85–1035.37

Table 2Bacterial identification, minimum inhibitory concentration (mg L⁻¹) and drought resistance of bacterial isolates (n = 3).

Strain	Closest relative sequence	% of similarity ^a	Cu	Pb	Cr	Cd	Ni	PEG6000, % (Mpa) ^b
TR15a	<i>Pseudomonas</i> sp.	99%	1750	3000	1000	750	1750	25 (–0.75)
TR15b	<i>Pseudomonas</i> sp.	99%	1750	1750	1000	300	1000	25 (–0.75)
TR15c	<i>Bacillus aerophilus</i>	99%	1750	3500	1000	750	600	25 (–0.75)
TR15e	<i>Stenotrophomonas rhizophila</i>	99%	1750	3000	1000	300	2250	15 (–0.30)
TR15a + TR15c	<i>Pseudomonas</i> sp. + <i>Bacillus aerophilus</i>	–	1750	3000	1000	750	1250	25 (–0.75)

^a Similarity with the strains of NCBI database.^b PEG6000 tolerance level.

1750 and 3500 mg L⁻¹ for Pb, 300–750 mg L⁻¹ for Cd and 600–2250 mg L⁻¹ for Ni. Except strain TR15e, all other tested strains showed moderate-to-high drought stress resistance under PEG6000. The strain TR15a showed the best resistance against most of the studied metals (Table 2). Since, gram positive bacteria are more sensitive to toxic metals, a decrease in Ni tolerance was observed in TR15c (Seiler and Berendonk, 2012). Two best strains were selected for preparation of consortium after analyzing MIC, drought tolerance, PGP attributes, antibiotic resistance of all four strains and separate experiments were conducted to check the efficiency of consortia TR15a + TR15c for above stated parameters. Consortium showed similar resistance potential for most of the studied metals and drought stress without showing any decrease in tolerance level with improved Ni resistance as compared to one of its constituent PGPRs TR15c.

The rhizospheric bacteria face much more competition, environmental stresses and pathogenic attacks than endophytic bacteria and thus antibiotics tolerance could provide a much better resistance and survival strength. All four individual strains and the consortium showed resistance against majority of the antibiotics (ampicillin: 10 μ g, kanamycin: 30 μ g, chloramphenicol: 30 μ g, penicillin: 6 μ g, tetracycline: 30 μ g and streptomycin: 30 μ g) (Table 3; Fig. 1). Except for tetracycline, TR15a showed resistance against all studied antibiotics whereas TR15c showed resistance against ampicillin, penicillin, tetracycline and kanamycin. The TR15e showed intermediate to susceptible response against almost all antibiotics. Because of improved resistance against chloramphenicol by the consortium TR15a + TR15c along with other

antibiotics except streptomycin and it could provide better resistance against rhizospheric infections as compared to individual PGPRs. Increase in Ni tolerance could be the possible reason for improved antibiotic resistance. Wani et al. (2009) and Ma et al. (2015) had also reported that antibiotic resistance properties of bacteria are often concomitant with heavy metal tolerance which is similar in present case.

3.4. Plant growth promoting attributes

The bacteria associated with the rhizosphere of metal tolerant plants exhibit different PGP characteristics such as ACC deaminase, siderophore production, IAA 1production, phosphate solubilization along with organic acids, ammonia and hydrogen cyanide which alleviates the metal stress and helps in plant growth (Singh et al., 2015; Gange and Gadhav, 2018; Grobelak et al., 2018). All the individual PGPRs showed significant concentration of ACC deaminase and in particular TR15a and TR15c exhibited high ACC deaminase of 44782 and 1560 nM α -ketobutyrate mg⁻¹ protein h⁻¹, respectively (Table 4). However, the maximum ACC deaminase activity was observed in the case of consortia of TR15a + TR15c and was 17%, 99%, 97% and 99% times higher than TR15a, TR15b, TR15c and TR15e, respectively. The deaminase enzyme metabolizes the ethylene precursor ACC which attenuate ethylene-mediated plant growth inhibition (Glick, 2014).

Siderophore an important biological process which assist to chelate Fe along with other metals can help to improve the nutrient uptake and sequester higher concentration of metal in plants. In

Table 3Antibiotic resistance of isolated bacterial strains (Mean $\pm$ SD; n = 3).

Bacterial strain	Ampicillin (10 μ g)	Penicillin (6 μ g)	Streptomycin (30 μ g)	Chloramphenicol (30 μ g)	Tetracycline (30 μ g)	Kanamycin (30 μ g)
TR15a	7.33 $\pm$ 0.58 (R)	14.67 $\pm$ 1.15 (R)	5.67 $\pm$ 1.53 (R)	0 (R)	21.67 $\pm$ 1.15 (S)	0 (R)
TR15b	0 (R)	12.67 $\pm$ 1.15 (R)	24.67 $\pm$ 3.21 (S)	12.67 $\pm$ 1.15 (R,S)	8.33 $\pm$ 0.58 (R)	18.33 $\pm$ 1.53 (S)
TR15c	0.33 $\pm$ 0.58 (R)	0 (R)	25.33 $\pm$ 1.53 (S)	23.33 $\pm$ 2.52 (S)	13.33 $\pm$ 2.08 (R)	10.00 $\pm$ 1.0 (R)
TR15e	31.67 $\pm$ 3.06 (S)	29.33 $\pm$ 2.52 (S)	21.33 $\pm$ 2.52 (S)	13.67 $\pm$ 1.53 (I)	27.00 $\pm$ 2.0 (S)	17.67 $\pm$ 0.58 (I,S)
TR15a + TR15c	6.45 $\pm$ 2.85 (R)	13.89 $\pm$ 1.88 (R)	15.45 $\pm$ 2.08 (S)	11.89 $\pm$ 2.72 (R)	13.45 $\pm$ 1.28 (R)	7.25 $\pm$ 2.54 (R)

Reported values are diameter of the inhibition zone (in mm); (R): resistant; (I): intermediate; (S): susceptible.

Ampicillin: $\leq$ 13 mm: resistant, 14–16 mm: intermediate, $\geq$ 17 mm: susceptible; Penicillin: $<$ 15 mm: resistant, $>$ 15 mm: susceptible.Streptomycin: $\leq$ 11 mm: resistant, 12–14 mm: intermediate, $\geq$ 15 mm: susceptible; Chloramphenicol: $\leq$ 12 mm: resistant, 13–17 mm: intermediate, $\geq$ 18 mm: susceptible;Tetracycline: $\leq$ 14 mm: resistant, 15–18 mm: intermediate, $\geq$ 19 mm: susceptible; Kanamycin: $\leq$ 13 mm: resistant, 14–17 mm: intermediate, $\geq$ 18 mm: susceptible.

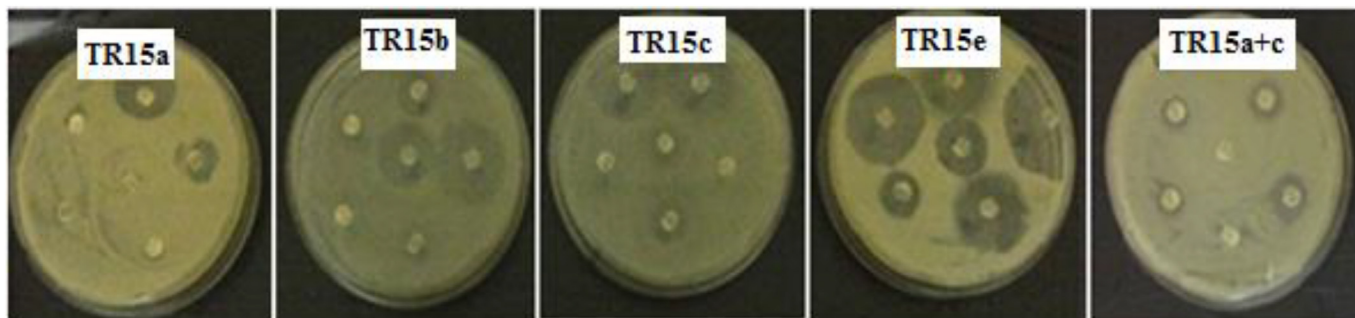


Fig. 1. Zone of inhibition under different tested antibiotics. Details about tested antibiotics are listed in Table 3.

Table 4

Plant growth promoting attributes of isolated bacterial strains (Mean $\pm$ SD; n = 3).

Bacterial strain	ACC deaminase activity (nM α -ketobutyrate mg^{-1} protein h^{-1})	Siderophore production zone: colony (mm)	Phosphate solubilization (mg L^{-1})	IAA production (mg L^{-1})	Ammonia production	HCN production
TR15a	44,782 $\pm$ 5280 b	Ns	297.63 $\pm$ 14.00 ab	5.32 $\pm$ 0.45 b	++	–
TR15b	340 $\pm$ 57 de	Ns	239.47 $\pm$ 17.65 c	3.35 $\pm$ 0.25 cd	++	–
TR15c	1,560 $\pm$ 245 c	12:6	10.50 $\pm$ 2.54 d	4.35 $\pm$ 0.65 c	++	–
TR15e	593 $\pm$ 65 d	Ns	271.32 $\pm$ 10.50 b	5.32 $\pm$ 0.78 b	+	–
TR15a + TR15c	53,745 $\pm$ 4542 a	12:7	315.54 $\pm$ 7.55 a	7.92 $\pm$ 0.48 a	++	–

++: reddish brown; +: brown; -:negative; Ns: no siderophore production.

Different alphabetical letters in the same column indicate significant difference at $p < 0.05$ level of significance.

addition, siderophore producing PGPRs also evidenced their potential role in Cu sequestration (Dimkpa et al., 2009; Rajkumar et al., 2010; Hesse et al., 2018) and hence played a vital role in the present study. The CAS agar plates inoculated with PGPRs showed orange-yellow zone around the inoculated colony which confirms its siderophore production. Only one strain TR15c (out of four selected PGPRs) showed its potential to produce siderophore however, TR15c containing consortia also showed its evidence in siderophore production.

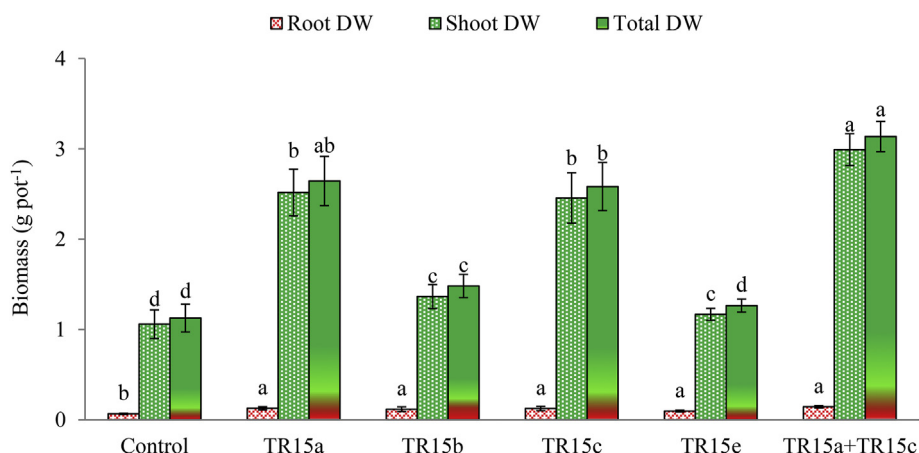
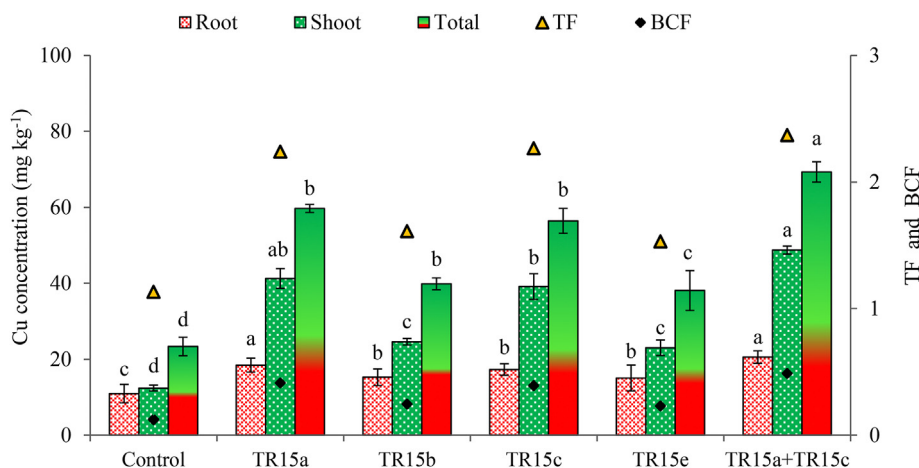
In present research, quantitative analysis of all four PGPRs showed IAA production (3.35–5.32 mg L^{-1}) and phosphate solubilization (10–297 mg L^{-1}) however, they were significantly ($p < 0.05$) lower than consortium TR15a + TR15c (IAA: 7.92 and P: 315 mg L^{-1}) which showed enhanced potential of consortia due to united effect of both PGPRs (Table 4). It is well documented that IAA helps in differentiation and cell division of the root and hence promotes root hair which provides nutrient and water uptake and chances for colonization of higher number of rhizobacteria. Etesami et al. (2015) had also reported that maximum correlation for the root growth in rice plant was mainly related to IAA followed by ACC deaminase activity and siderophore production. Moreover, all four strains including consortia showed solubilization of inorganic tricalcium phosphate. Production of organic acid during phosphate solubilization could further help in lowering of pH and promotes accumulation of metals and nutrients from the soil for improved plant growth (Tripti et al., 2017). In addition, both individual and consortia also showed positive and negative results for ammonia and HCN production, respectively. Studies have evidenced that inoculation of PGPRs in the form of consortium could provide synergistic effect on the growth of plants, disease suppression and metal accumulation (Dary et al., 2010; Kumar et al., 2018) however, still there is limited number of research done and there is a great need to foster plant-microbe consortium studies.

3.5. Seed germination and biometric growth parameters

The effect of four PGPRs and consortia (TR15a + TR15c) on seed germination and growth of *H. annuus* in comparison to uninoculated control are presented in Table 5. Application of PGPRs significantly ($p < 0.05$) improved the percentage of seed germination and was 24%, 20%, 20% and 11% for TR15a, TR15b, TR15c and TR15e higher than uninoculated control, respectively. Whereas overall, maximum percentage of seed germination of 96% was observed in case of consortium which showed 33% higher seed germination than uninoculated control. Sunflower plants showed its potential to tolerate 100 mg Cu kg^{-1} without showing any necrosis and chlorosis. Application of PGPRs also resulted in improved root and shoot length and biomass of the plant. The roots of TR15e, TR15b, TR15c and TR15a and TR15a+15c were 12%, 26%, 27%, 32%, and 42% higher than uninoculated control, respectively. Similarly, shoot also showed the similar trend of aerial part elongation. In case of biomass the dry weight of root and shoots of TR15a and TR15c treated plants showed enhanced biomass than uninoculated control however, their consortium produced supreme effect than any other treatment and showed a maximum increase of 54% and 65% for root and shoot over uninoculated control, respectively. It was found that application of consortium TR15a + TR15c produced significantly ($p < 0.05$) better effect for root and shoot length, fresh and dry biomass of root and shoot and total biomass than individual PGPRs treated plants (Fig. 2). Studies have evidenced that application of metal tolerant, IAA, siderophore producing, ACC deaminase producing and phosphate solubilizing rhizospheric bacteria plays a vital role in elongation of root and shoot and overall growth of the plant by alleviating the metal toxicity and other environmental stresses (Glick 2014; Kumar and Tripti 2018). The vigor index (percentage of seed germination $\times$ total length of seedling shoot and root) (Ma et al., 2016) of consortium was 2.65-fold higher than control and between 1.34 and 2.01 fold higher than four studied individual PGPRs, respectively.

Table 5Effect of Cu-resistant single and consortium PGPR on vigour index, length and fresh biomass of root and shoot of sunflower grown in Cu amended soil. Mean $\pm$ SD; n = 4.

Bacterial strain	Seed germination (%)	Vigor index (VI%)	Length (cm)		Fresh biomass (g)	
			Root	Shoot	Root	Shoot
Control	64	461 de	6.50 $\pm$ 0.41 de	33.00 $\pm$ 2.45 e	0.70 $\pm$ 0.08 d	4.77 $\pm$ 0.36 de
TR15a	84	907 b	9.50 $\pm$ 0.71 b	49.25 $\pm$ 3.30 b	1.29 $\pm$ 0.08 b	8.62 $\pm$ 0.36 b
TR15b	80	792 c	8.75 $\pm$ 0.48 c	39.00 $\pm$ 2.45 d	1.16 $\pm$ 0.08 bc	7.07 $\pm$ 0.16 c
TR15c	80	807 c	8.87 $\pm$ 0.48 c	44.00 $\pm$ 3.37 c	1.21 $\pm$ 0.11 b	7.92 $\pm$ 0.60 b
TR15e	72	607 d	7.37 $\pm$ 0.75 d	38.75 $\pm$ 2.06 d	1.06 $\pm$ 0.10 c	5.24 $\pm$ 0.45 d
TR15a + TR15c	96	1223 a	11.25 $\pm$ 1.04 a	59.00 $\pm$ 2.94 a	1.49 $\pm$ 0.12 a	10.93 $\pm$ 0.99 a

Different alphabetical letters in the same column indicate significant difference at $p < 0.05$ level of significance.**Fig. 2.** Influence of Cu-resistant single and consortium PGPR on shoot, root and total dry weight (DW) of sunflower plant grown in Cu treated soil (Mean $\pm$ SD; n = 4). Different alphabetical letters indicate significant difference at $p < 0.05$ level of significance.**Fig. 3.** Root shoot and total Cu uptake in sunflower plant with, without and consortium of PGPRs (Mean $\pm$ SD; n = 4); Translocation factor (TF): ratio of metal in shoot to metal in root; Bioconcentration factor (BCF): ratio of metal in shoot to metal in soil.

3.6. Accumulation of Cu in sunflower and metal in post-harvest soil

The effects of Cu accumulation in below and above ground tissues of sunflower and its transfer from soil to root to shoot are presented in Fig. 3. Sunflower showed its potential to tolerate critical soil total Cu concentration of 100 mg kg⁻¹ without showing any toxicity symptoms. The application of PGPRs improved the Cu extraction from the soil and its transfer from root to shoot as compared to uninoculated control. Significantly ($p < 0.05$), maximum accumulation of Cu was observed in consortia inoculated sunflower and was 47% and 75% higher for root and shoot,

respectively as compared to uninoculated control. Our results were in accordance with the findings of Carrillo-Castaneda et al. (2005), Zhang et al. (2011) and Singh and Jha (2018), who also reported that ACC deaminase and siderophore producing bacteria can help in enhanced growth and consequently Cu uptake in *Brassica napus*, *Phaseolus vulgaris* and *Triticum aestivum*, respectively.

In addition, phosphate solubilization causes decrease in the pH of the soil and thus enhanced metal bioavailability in the rhizosphere which further helped siderophore to chelates with the Fe and Cu and helped in its absorption through roots (Dimkpa et al., 2009; Rajkumar et al., 2009; Hesse et al., 2018). ACC deaminase

along with IAA help to alleviate both biotic and abiotic stress factors such as drought, flower wilting, temperature, salinity, metal, bacterial and fungal infections and can play important role in delay in senescence (Kumar et al., 2012; Glick 2014; Tripti and Kumar et al., 2018).

Present comparative study of consortium (TR15a + TR15c) with individual PGPRs showed that high ACC deaminase producing strain TR15a and siderophore producing strain TR15c could provide synergistic effect on Cu accumulation in sunflower plant with improved translocation factor and bioaccumulation factors. A successful plant growth promoting inoculants are those inoculants which colonize the root system during the growing season (Zhang et al., 2011). The population of $4.55\text{--}6.78 \times 10^4$ cfu g⁻¹ in the fresh rhizospheric soil of PGPRs inoculated pots ensures their viability, ampicillin resistance and colonization under Cu stress. Post-harvest analysis of soil suggests maximum Cu removal by TR15a + TR15c > TR15a > TR15c > TR15b > and TR15e which was 51%, 40%, 37%, 25% and 23%, respectively. In order to increase the effectiveness of phytoremediation potential, multiple cropping is required.

4. Conclusions

Our study suggests that *Trifolium repens* growing in heavily contaminated Cu contaminated site harbours significant number of high metal tolerant and drought resistant rhizobacteria with multiple plant growth promoting attributes namely IAA, siderophore, ACC deaminase, ammonia, and phosphate solubilization. The strains *Pseudomonas* sp. TR15a, *Bacillus aerophilus* TR15c and their consortium showed higher resistance to various heavy metals (including Cu) and antibiotics. Moreover, consortia of TR15a + TR15c showed improved results for IAA production and phosphate solubilization than individual PGPR isolates. Consortia of high ACC deaminase producing TR15a and siderophore producing TR15c showed better Cu accumulation than individual bacterial isolates without compromising the biomass of sunflower plant. Since, consortium studies are limited but provides better results, the synergistic activity of the members of the investigated consortium for the effective accumulation of Cu provides valuable information for its further application in bioremediation of Cu contamination in natural ecosystem. Further, it will help to explore the molecular mechanisms (gene expression level) involved in microbe mediated alteration of Cu uptake in plants.

CRediT author statement

Adarsh Kumar: Conceptualization, Formal analysis, Methodology, Investigation, Validation, Writing - original draft, Writing - review & editing. Tripti: Conceptualization, Investigation, Methodology, Writing - original draft, Writing - review & editing, Supervision, Project administration, Funding acquisition. Maria Maleva: Resources, Formal analysis, Visualization, Project administration, Writing - review & editing. Mani Rajkumar: Visualization, Funding acquisition, Writing - review & editing. L. Benedict Bruno: Writing - review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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